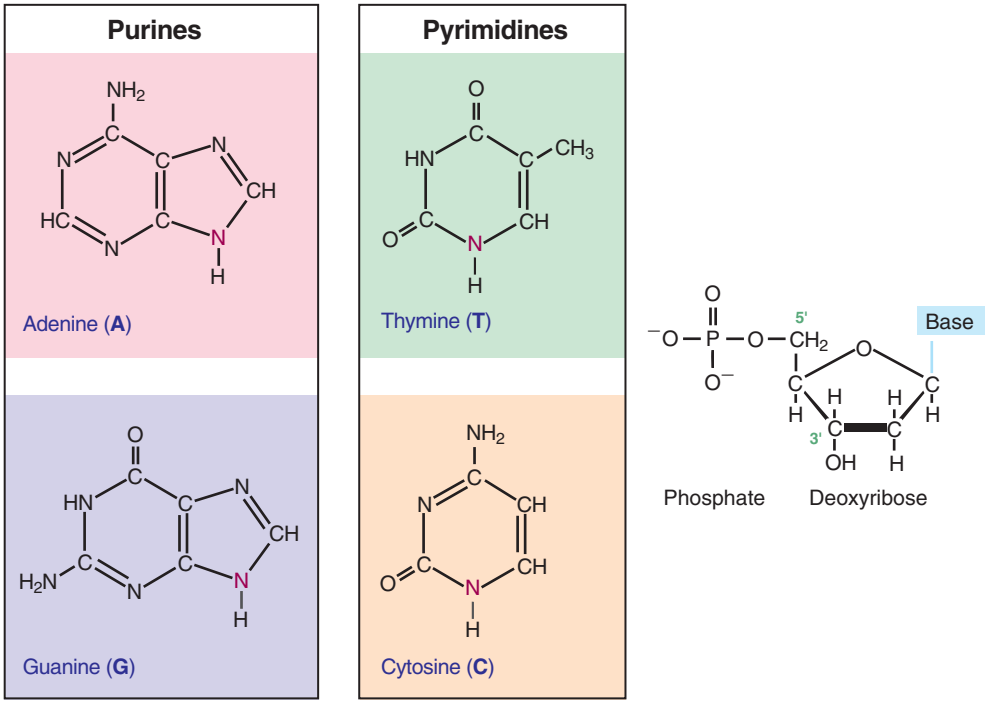


Figure 2-2 The four bases of DNA and the general structure of a nucleotide in DNA. Each of the four bases bonds with deoxyribose (through the nitrogen shown in *magenta*) and a phosphate group to form the corresponding nucleotides.

GENES IN THE HUMAN GENOME

What is a gene? And how many genes do we have? These questions are more difficult to answer than it might seem.

The word *gene*, first introduced in 1908, has been used in many different contexts since the essential features of heritable "unit characters" were first outlined by Mendel over 150 years ago. To physicians (and indeed to Mendel and other early geneticists), a gene can be defined by its observable impact on an organism and on its statistically determined transmission from generation to generation. To medical geneticists, a gene is recognized clinically in the context of an observable variant that leads to a characteristic clinical disorder, and today we recognize approximately 5000 such conditions (see Chapter 7).

The Human Genome Project provided a more systematic basis for delineating human genes, relying on DNA sequence analysis rather than clinical acumen and family studies alone; indeed, this was one of the most compelling rationales for initiating the project in the late 1980s. However, even with the finished sequence product in 2003, it was apparent that our ability to recognize features of the sequence that point to the existence or identity of a gene was sorely lacking. Interpreting the human genome sequence and relating its variation to human biology in both health and disease is thus an ongoing challenge for biomedical research.

Although the ultimate catalogue of human genes remains an elusive target, we recognize two general types of gene, those whose product is a protein and those whose product is a functional RNA.

- The number of **protein-coding genes**—recognized by features in the genome that will be discussed in Chapter 3—is estimated to be somewhere between 20,000 and 25,000. In this book, we typically use approximately 20,000 as the number, and the reader should recognize that this is both imprecise and perhaps an underestimate.
- In addition, however, it has been clear for several decades that the ultimate product of some genes is not a protein at all but rather an RNA transcribed from the DNA sequence. There are many different types of such RNA genes (typically called **noncoding genes** to distinguish them from protein-coding genes), and it is currently estimated that there are at least another 20,000 to 25,000 noncoding RNA genes around the human genome.

Thus overall—and depending on what one means by the term—the total number of genes in the human genome is of the order of approximately 20,000 to 50,000. However, the reader will appreciate that this remains a moving target, subject to evolving definitions, increases in technological capabilities and analytical precision, advances in informatics and digital medicine, and more complete genome annotation.

DNA Structure: A Brief Review

Before the organization of the human genome and its chromosomes is considered in detail, it is necessary to review the nature of the DNA that makes up the genome. DNA is a polymeric nucleic acid macromolecule composed of three types of units: a five-carbon sugar, deoxyribose; a nitrogen-containing base; and a phosphate group (Fig. 2-2). The bases are of two types, **purines** and **pyrimidines**. In DNA, there are two purine bases, **adenine** (A) and **guanine** (G), and two pyrimidine